

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	Program	Master of Computer Applications - Artificial Intelligence & Machine Learning(MC306)
Master Subject Coordinator Name:	Sweta	Master Subject Coordinator E-Code:	E15013
Course Name	Artificial Intelligence Tools -I	Course Code	25CAP-614

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
0	0	4	0	2.00	P

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Practical Examination (PRAC)	Practical (PRAC)

Mission of the Department	M1 To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2 To provide a frame work through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4 To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5 To undertake societal activities for upliftment of rural/deprived sections of society
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)

PEO1	Demonstrate analytical and design skills including the ability to generate creative solutions and foster team-oriented professionalism through effective communication in their careers.
PEO2	Expertise in successful careers based on their understanding of formal and practical methods of application development using the concept of computer programming languages and design principles in accordance to industry 4.0
PEO3	Exhibit the growth of the nation and society by implementing and acquiring knowledge of upliftment of health, safety and other societal issues.
PEO4	Implement their exhibiting critical thinking and problem- solving skills in professional practices or tackle social, technical and business challenges.

Program Specific OutComes(PSOs)

PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)

PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	4th Edition	Pearson	2020
2	Machine Learning: A Probabilistic Perspective	Kevin P. Murphy	2nd Edition	The MIT Press	2022

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Fundamentals of Machine Learning for Predictive Data Analytics	John D. Kelleher, Brian Mac Namee, and Aoife D'Arc	2nd Edition	The MIT Press	2020
2	Interpretable Machine Learning	Christoph Molnar	2nd Edition	Self-published	2022

Course OutCome	
SrNo	OutCome
CO1	Gain a foundational understanding of Artificial Intelligence (AI) concepts and applications
CO2	Analyze and formulate single-agent search problems in state space representation
CO3	Apply various search algorithms to solve problems using Python
CO4	Evaluate the efficiency and effectiveness of different search algorithms based on problem characteristics
CO5	Developing AI agents that can play games by utilizing search algorithms to make optimal moves

Lecture Plan Preview-Practical					
Unit No	ExperimentNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer(s)
1	1	1(a)Implement Breadth-First Search (BFS) Algorithm: Introduction to Breadth-First Search (BFS) and Algorithm, Pseudo code.	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO1,CO2

1	2	1(b)Implement Breadth-First Search (BFS) Algorithm: Python Implementation – Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO3
1	3	1(c)Implement Breadth-First Search (BFS) Algorithm: Test Cases and Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO3
1	4	2(a) Implement Depth-First Search (DFS) Algorithm: Introduction to Depth-First Search (DFS) Algorithm and Pseudocode	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO1,CO3
1	5	2(b) Implement Depth-First Search (DFS) Algorithm: Python Implementation – Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO3
1	6	2(c)Implement Depth-First Search (DFS) Algorithm: Test Cases and Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO3
2	7	3(a)Implement Best-First Search Algorithm with a chosen heuristic: Introduction to Best-First Search (Greedy) Algorithm and Heuristic Function	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4

2	8	3(b)Implement Best-First Search Algorithm with a chosen heuristic: Python Implementation – Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4
2	9	3(c)Implement Best-First Search Algorithm with a chosen heuristic: Test Cases and Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4
2	10	4(a)Implement A* Search Algorithm: Introduction to A* Algorithm Mechanics and Cost Functions	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4
2	11	4(b)Implement A* Search Algorithm: Python Implementation – Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4
2	12	4(c)Implement A* Search Algorithm: Test Cases and Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO2,CO4
3	13	5(a)Implementing Hill Climbing: Introduction to Hill Climbing and State-Space, Algorithm and Variants	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO4

3	14	5(b)Implementing Hill Climbing: Python Implementation – Python Implementation – Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO4
3	15	5(c)Implementing Hill Climbing: Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO4
3	16	6(a)Implement Minimax Algorithm: Introduction to Game Trees and Minimax Algorithm and Pseudocode	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO5
3	17	6(b)Implement Minimax Algorithm: Python Implementation - Step-by-Step	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO5
3	18	6(c)Implement Minimax Algorithm: Visualization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO3,CO5
3	19	Project 1 (Phase1:Project title)Smart Path finding in a Dynamic Grid World	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	20	Project 1(Phase2: Design and Implementation)Smart Path finding in a Dynamic Grid World	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5

3	21	Project 1(Phase3: Online Platform)Smart Path finding in a Dynamic Grid World	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	22	Project 2(Phase1:Project title) Delivery Route Optimization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	23	Project 2(Phase2: Design and Implementation) Delivery Route Optimization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	24	Project 2(Phase3: Online Platform) Delivery Route Optimization	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	25	Project 3 (Phase1:Project title) Game AI for Tic-Tac- Toe	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	26	Project 3(Phase2: Design and Implementation) Game AI for Tic-Tac- Toe	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	27	Project 3(Phase2: Design and Implementation) Game AI for Tic-Tac- Toe	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5

3	28	Project 4(Phase1:Project title) Treasure Hunt Game Agent	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	29	Project 4(Phase2: Design and Implementation)Treas ure Hunt Game Agent	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5
3	30	Project 4(Phase3: Online Platform) Treasure Hunt Game Agent	,T-Artificial Intelligence: A Mod,T-Machine Learning: A Probabilis,R- Interpretable Machine Learnin,R-Fundamentals of Machine Learni	Hand On Activity based,Infographics Practical,Instructor Lead Workshop Practical,Professor of Practice/Adjunct Faculty/Visiting Professors,Simulati on Practical,Video Demonstration	CO4,CO5

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	20PRAC01	External Viva / Voce	40
2	20PRAC01	Experiment-1	30
3	20PRAC01	Experiment-2	30
4	20PRAC01	Experiment-3	30
5	20PRAC01	Experiment-4	30
6	20PRAC01	Experiment-5	30
7	20PRAC01	Experiment-6	30
8	20PRAC01	Experiment-7	30
9	20PRAC01	Experiment-8	30
10	20PRAC01	Experiment-9	30
11	20PRAC01	Experiment-10	30
12	20PRAC01	Mid-Term Test	15

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	NA	NA	1	NA	2	NA	NA	NA	NA	NA	2	2
CO2	3	2	3	3	2	NA	NA	NA	NA	NA	NA	NA	2	3
CO3	2	NA	3	NA	NA	NA	3	NA	NA	NA	NA	NA	3	2
CO4	3	3	NA	1	3	NA	NA	NA	NA	NA	NA	NA	3	3
CO5	3	NA	3	2	3	NA	2	NA	NA	NA	NA	NA	1	2
Target	2.8	2.33	3	2	2.25	NA	2.33	NA	NA	NA	NA	NA	2.2	2.4

